SOFT COMPUTING (CSC425)

DIMENSIONALITY REDUCTION

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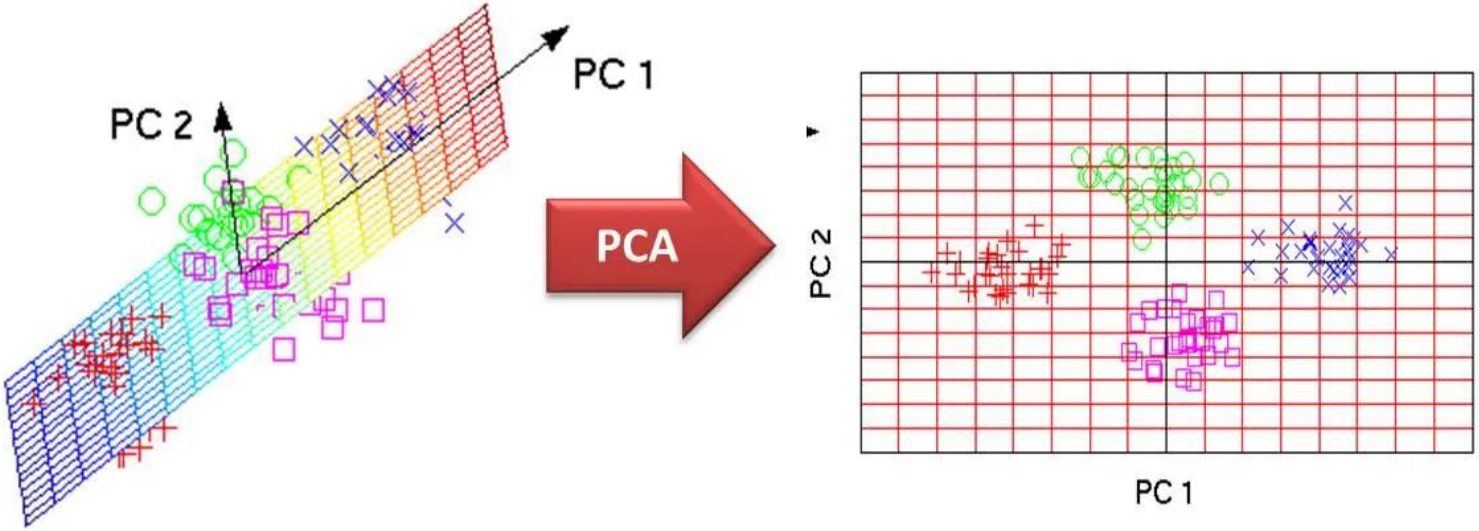
Karhunen Loeve Transform (KLT)

Hotelling Transform

Principal Components Analysis (PCA)

Dimensionality Reduction
&
Principal Component Analysis



PRINCIPAL COMPONENT ANALYSIS (PCA)

- Principal Component Analysis is an **unsupervised learning** algorithm that is used for the dimensionality reduction in machine learning.
- It is a statistical process that converts the observations of correlated features into a set of linearly uncorrelated features with the help of orthogonal transformation.
- These new transformed features are called the **Principal Components**.
- It is one of the popular tools that is used for exploratory data analysis and predictive modeling.
- PCA generally tries to find the lower-dimensional surface to project the high-dimensional data.

PRINCIPAL COMPONENTS ANALYSIS (PCA)

Can be used to:

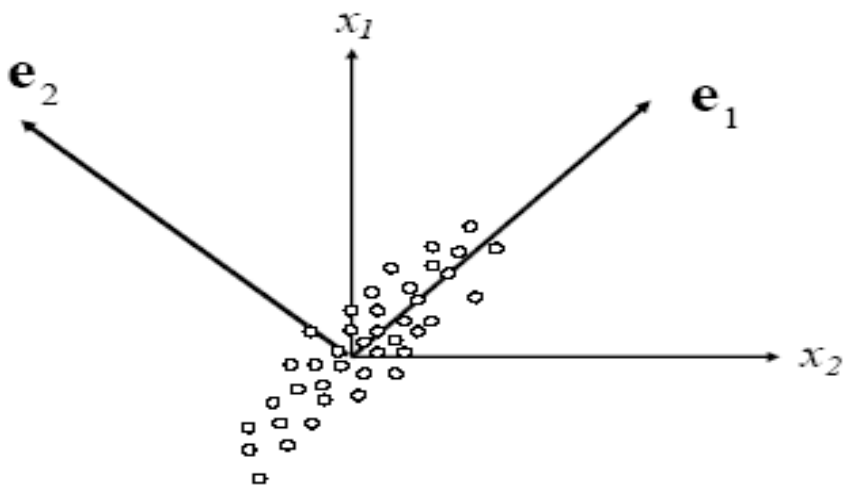
- Reduce number of dimensions in data
- Find patterns in high-dimensional data
- Visualize data of high dimensionality

Example applications:

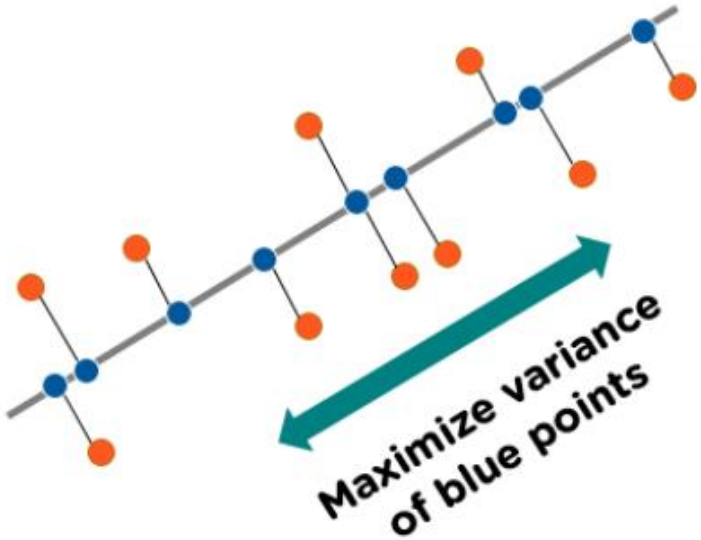
- Face recognition
- Image compression
- Gene expression analysis

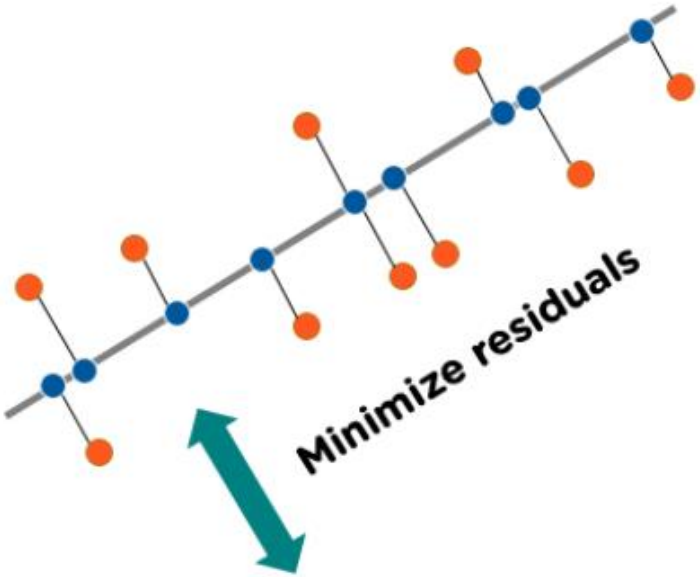
BASIC IDEA BEHIND PCA

PCA – Rotational Transform



HOW DOES PCA WORK?

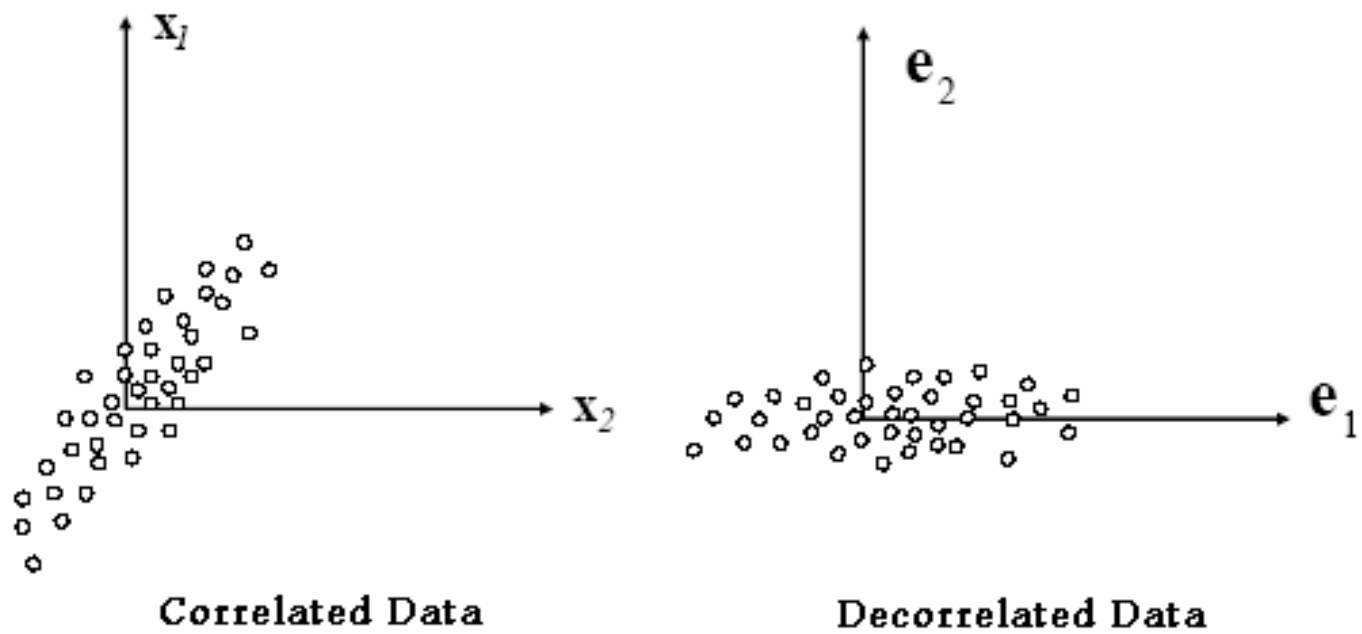




EIGENVALUE INDICATES THE VARIANCE

$$\lambda_i = \sigma^2_{\mathbf{e}_i}$$

CORRELATED TO UNCORRELATED DATA



BASIC STATISTICS

$$\overline{X} = \frac{\sum X}{n}$$

$$Var(X) = \frac{\sum_{i=1}^{n} \left(X - \overline{X}\right)^2}{n-1}$$

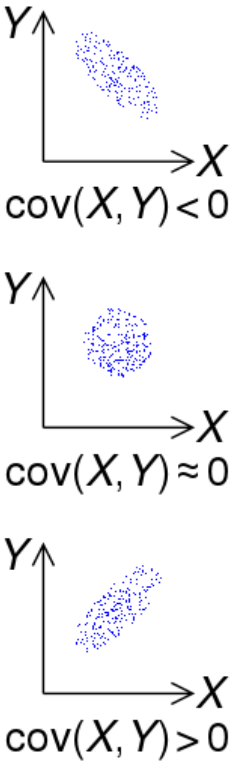
$$\text{Covar}(X,Y) = \frac{\sum_{i=1}^{n} \left(X - \overline{X}\right)\left(Y - \overline{Y}\right)}{n-1}$$

COVARIANCE MATRIX

How *two* dimensions vary from the mean with respect to ec

Example for three dimensions (x,y,z):

$$C = \begin{bmatrix} \mathrm{cov}(x,x) & \mathrm{cov}(x,y) & \mathrm{cov}(x,z) \\ \mathrm{cov}(y,x) & \mathrm{cov}(y,y) & \mathrm{cov}(y,z) \\ \mathrm{cov}(z,x) & \mathrm{cov}(z,y) & \mathrm{cov}(z,z) \end{bmatrix}$$



Cov(X, Y) > 0: Dimensions increase together

Cov(X, Y) < 0: One increases, one decreases

Cov(X, Y) = 0: Dimensions are independent

MATRIX ALGEBRA

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 11 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4 \times \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Example of one non-eigenvector and one eigenvector

MATRIX ALGEBRA

$$2 \times \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} 24 \\ 16 \end{bmatrix} = 4 \times \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

Example of how a scaled eigenvector is still an eigenvector

EIGENVECTORS PROPERTIES

Eigenvectors can only be found for square matrices and not every square matrix has eigenvectors.

Given an *n x n matrix that does have eigenvectors, there are n of them*.

Another property of eigenvectors is that even if we scale the vector by some amount before we multiply it, we still get the same multiple of it as a result. This is because if you scale a vector by some amount, all you are doing is ***making it longer not changing it's direction.***

Lastly, all the eigenvectors of a matrix are *perpendicular. It means that you can express the data in terms of these perpendicular eigenvectors, instead of expressing them in terms of the x and y axes.*

EIGENVALUES

Eigenvalues is the amount by which the original vector was scaled after multiplication by the square matrix.

4 value is the *eigenvalue associated with that eigenvector in the example*.

No matter what multiple of the eigenvector we took before we multiplied it by the square matrix, we would always get 4 times the scaled vector.

Eigenvectors and Eigenvalues always come in pairs.

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4 \times \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

EIGENVALUES & EIGENVECTORS CALCULATIONS

Vectors x having same direction as Ax are called *eigenvectors* of A (A is an n by n matrix).

In the equation $Ax=\lambda x$, λ is called an *eigenvalue* of A.

$Ax=\lambda x \Leftrightarrow (A-\lambda I)x=0$

How to calculate x and λ:

- Calculate $det(A-\lambda I)$, yields a polynomial (degree n)
- Determine roots to $det(A-\lambda I)=0$, **roots are eigenvalues λ**
- Solve $(A-\lambda I)\,\mathbf{x}=0$ for each λ to obtain eigenvectors **x**

DEFINITION OF EIGENVECTORS & EIGENVALUES

Definition: The *eigenvalues* of a real matrix **M** are the real numbers λ for which there is a nonzero vector **e** such that

$$\boxed{\mathbf{Me} = \lambda\, \mathbf{e}.}$$

The *eigenvectors* of **M** are the nonzero vectors **e** for which there is a real number λ such that $\mathbf{Me} = \lambda\, \mathbf{e}$.

If $\mathbf{Me} = \lambda\, \mathbf{e}$ for $\mathbf{e} \neq 0$, then **e** is an *eigenvector* of **M** associated with *eigenvalue* λ, and vice versa. The eigenvectors and corresponding eigenvalues of **M** constitute the *eigensystem* of **M**.

DEFINITION OF EIGENVECTORS & EIGENVALUES

Example: Consider the matrix

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

It is easy to verify that $\mathbf{M}\mathbf{e}_1 = \lambda_1 \mathbf{e}_1$ and $\mathbf{M}\mathbf{e}_2 = \lambda_2 \mathbf{e}_2$ for $\lambda_1 = 1$, $\lambda_2 = 2$ and

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

In other words, $\mathbf{e}_1$ is an eigenvector of $\mathbf{M}$ with associated
eigenvalue λ_1, and similarly for $\mathbf{e}_2$ and λ_2.

EIGENVALUES

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Example
for Computation of Eigenvectors & Eigenvalues

EXAMPLE

$$[A]=\begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix} \qquad [A]-\lambda[I]=\begin{bmatrix} 3-\lambda & -1.5 \\ -0.75 & 0.75-\lambda \end{bmatrix}$$

$$\det([A]-\lambda[I]) = (3-\lambda)(0.75-\lambda)-(-0.75)(-1.5)=0$$

$$2.25-0.75\lambda-3\lambda+\lambda^2-1.125=0$$

$$\lambda^2-3.75\lambda+1.125=0$$

$$\lambda=\frac{-(-3.75)\pm\sqrt{(-3.75)^2-4(1)(1.125)}}{2(1)}$$

$$= \qquad \frac{3.75\pm 3.092}{2}$$

= 3.421,0.3288 So the eigenvalues are 3.421 and 0.3288.

FIND THE EIGENVECTORS OF A

$$[X] = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

be the eigenvector corresponding to

$\lambda_1 = 3.421$

Hence

$$([A] - \lambda_1 [I])[X] = 0$$

$$\left\{ \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix} - 3.421 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -0.421 & -1.5 \\ -0.75 & -2.671 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

FIND THE EIGENVECTORS OF A

If $x_1 = s$ Then $-0.421s-1.5x_2=0$

$x_2 = -0.2807\ s$

The eigenvector corresponding to $\lambda_1 = 3.421$ then is

$$[X] = \begin{bmatrix} s \\ -0.2807s \end{bmatrix} = s\begin{bmatrix} 1 \\ -0.2807 \end{bmatrix}.$$

The eigenvector corresponding to $\lambda_1 = 3.421$ is $\begin{bmatrix} 1 \\ -0.2807 \end{bmatrix}$

Similarly, the eigenvector corresponding to $\lambda_2 = 0.3288$ is

$$\begin{bmatrix} 1 \\ 1.781 \end{bmatrix}$$

HOMEWORK

Find the eigenvalues and eigenvectors of

$$[A] = \begin{bmatrix} 10 & 9 \\ 2 & 3 \end{bmatrix}$$

using the determinant method

$$\textit{Answer: } (12,1), \begin{bmatrix} 0.9762 \\ 0.2169 \end{bmatrix}, \begin{bmatrix} 0.8381 \\ -0.8381 \end{bmatrix}$$

PRINCIPAL COMPONENT ANALYSIS (PCA)

PCA is a way of identifying patterns in data, and expressing the data in such a way as to highlight their similarities and differences.

Since patterns in data can be hard to find in data of high dimension, where the luxury of graphical representation is not available, PCA is a powerful tool for analysing data.

The other main advantage of PCA is that once you have found these patterns in the data, and you compress the data, ie. by reducing the number of dimensions, without much loss of information. This technique used in image compression.

PCA –ORIGINAL AND ADJUSTED DATA

Data =

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

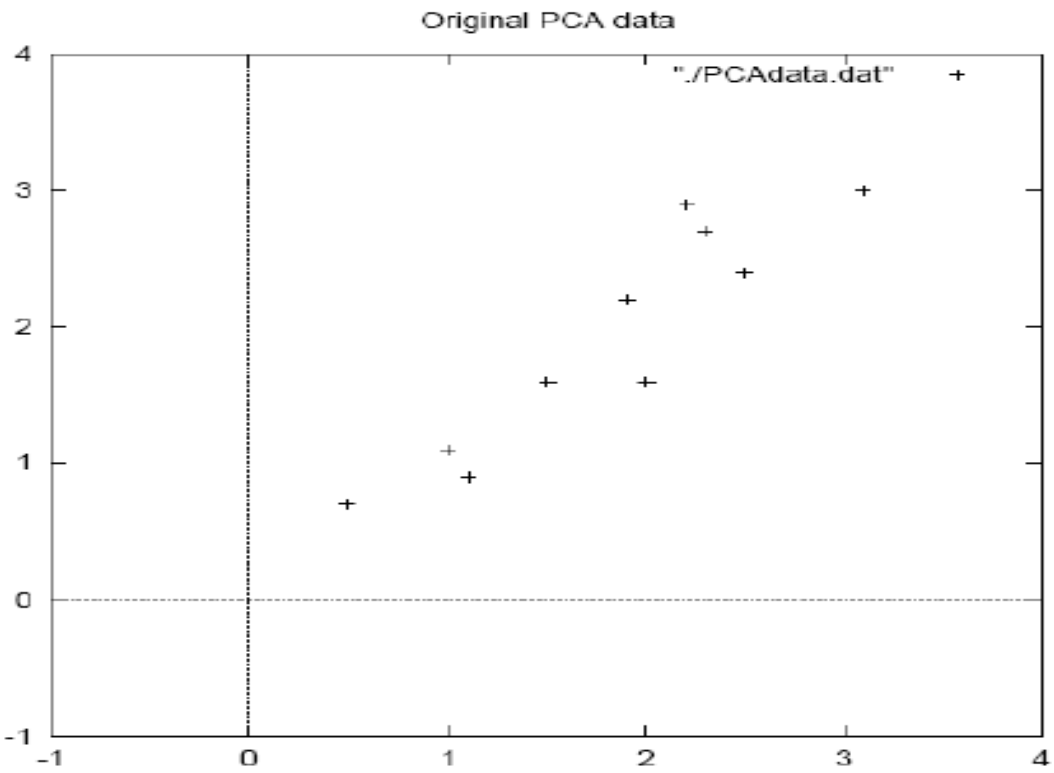
Original Data

DataAdjust =

x	y
.69	.49
-1.31	-1.21
.39	.99
.09	.29
1.29	1.09
.49	.79
.19	-.31
-.81	-.81
-.31	-.31
-.71	-1.01

Original Data - Average

PCA –ORIGINAL DATA PLOT



CALCULATE COVARIANCE MATRIX

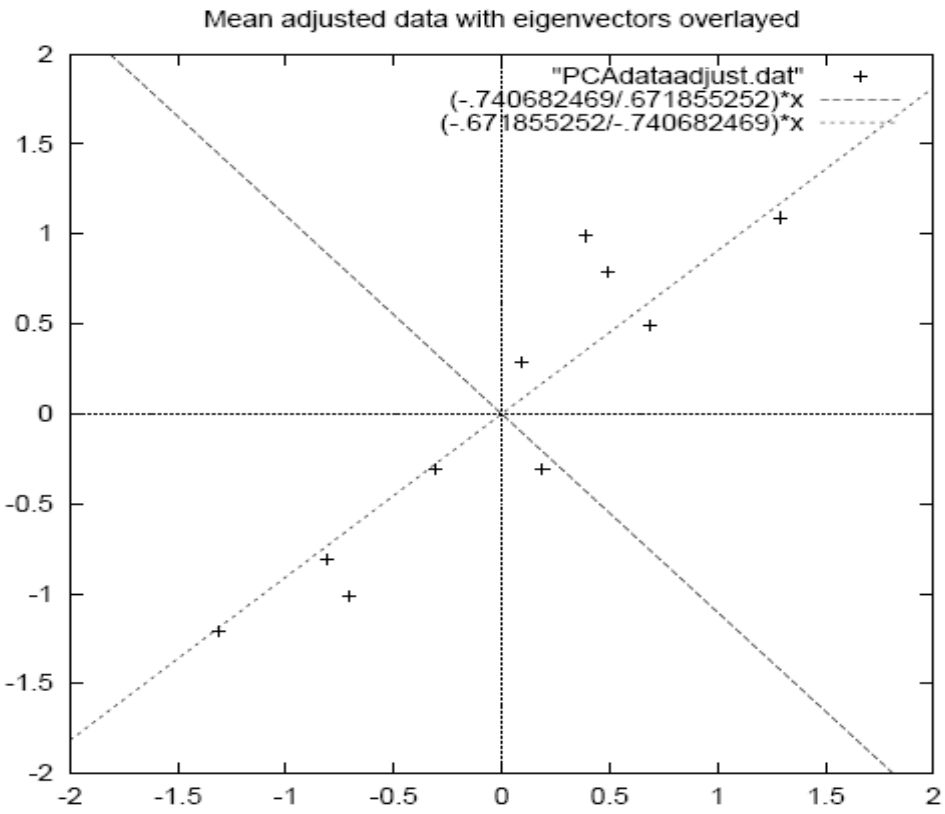
$$\text{cov} = \begin{bmatrix} 0.61656 & 0.61544 \\ 0.61544 & 0.71656 \end{bmatrix}$$

CALCULATE EIGENVECTORS AND EIGENVALUES FROM COVARIANCE MATRIX

$$eigenvalues = \begin{bmatrix} 0.049083 \\ 1.284028 \end{bmatrix}$$

$$eigenvectors = \begin{bmatrix} -0.73518 & -0.67787 \\ 0.67787 & -0.73518 \end{bmatrix}$$

EIGENVECTORS PLOT



CHOOSING COMPONENTS AND FORMING A FEATURE VECTOR

The eigenvector with the ***highest*** *eigenvalue is the* ***principle component*** *of the data set. It is the most significant relationship between the data dimensions.*

In general, once eigenvectors are found from the covariance matrix, the next step is to ***order them by eigen value****, highest to lowest. This gives us the components in order of significance.*

Now, if we like, we can decide to ***ignore*** *the components of lesser significance. We do lose some information, but if the eigenvalues are small, we don't lose much.*

If we leave out some components, the final data set will have less dimensions than the original.

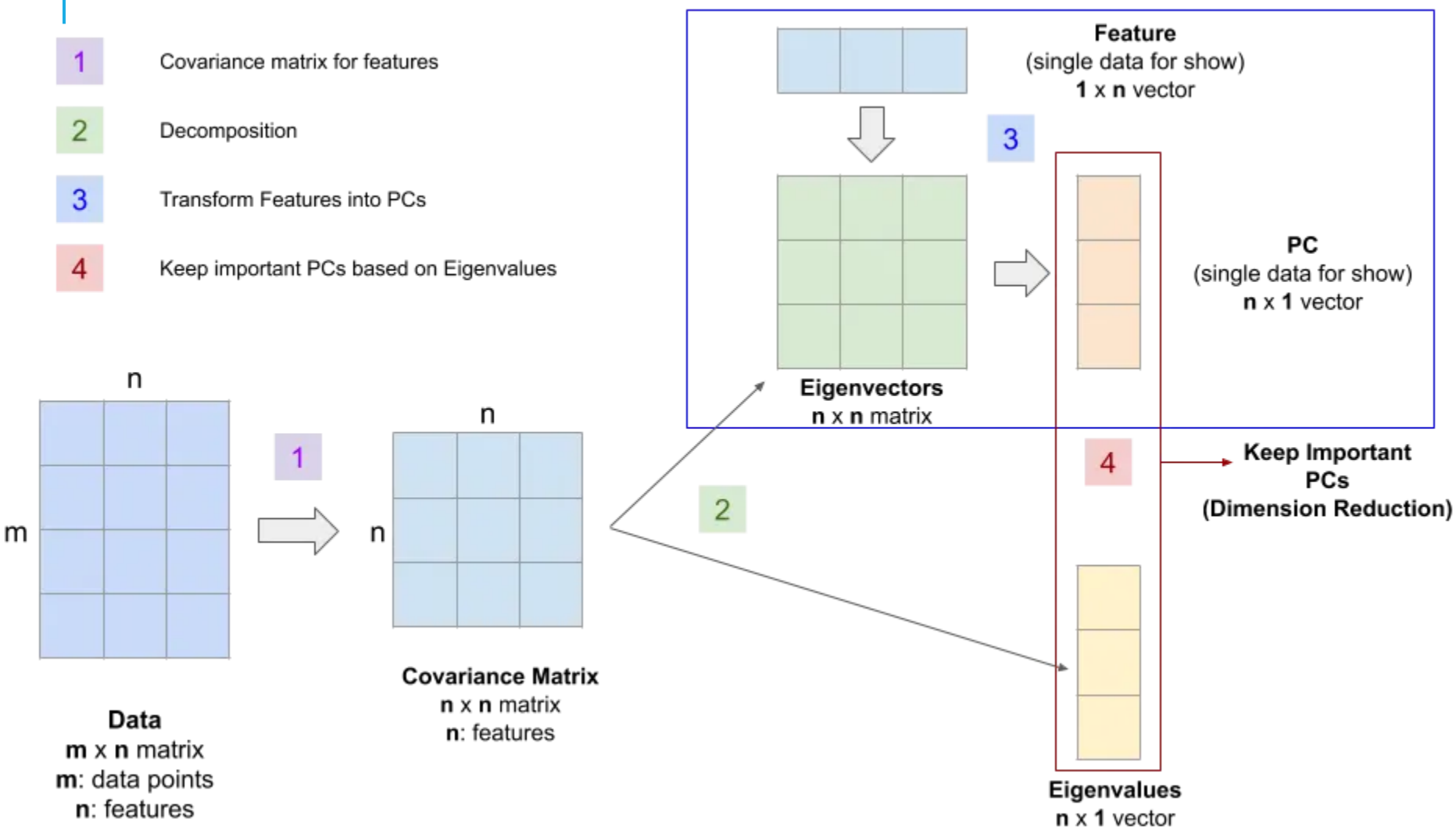
FEATURE VECTOR

$$FeatureVector = (eig_1, eig_2, eig_3 \ldots\ldots eig_n)$$

$$eigenvectors = \begin{bmatrix} -0.67787 & -0.73518 \\ -0.73518 & 0.677873 \end{bmatrix}$$

$$pc = \begin{bmatrix} -0.67787 \\ -0.73518 \end{bmatrix}$$

Steps of PCA algorithm


1
Covariance matrix for features
2
Decomposition
3
Transform Features into PCs
4
Keep important PCs based on Eigenvalues
Feature
(single data for show)
1 x n vector
3
PC
(single data for show)
n x 1 vector
Eigenvectors
n x n matrix
n
m
1
n
n
2
4
Keep Important
PCs
(Dimension Reduction)
Covariance Matrix
n x n matrix
n: features
Data
m x n matrix
m: data points
n: features
Eigenvalues
n x 1 vector

PCA – STEP 1

Get some data!

- The data is organized as a matrix *Data*
- Rows are observations
- Columns are dimensions (variables)
- Observations can also act as dimensions and vice versa by transposing matrix

PCA – STEP 2

Subtract the mean from each dimension

$$DataAdjust_{ij} = Data_{ij} - \overline{\{Data_{xj} \mid x = 1,\ldots,m\}}$$

where *i* is observation, *j* is dimension and *m* is total number of observations

PCA – STEPS 3 & 4

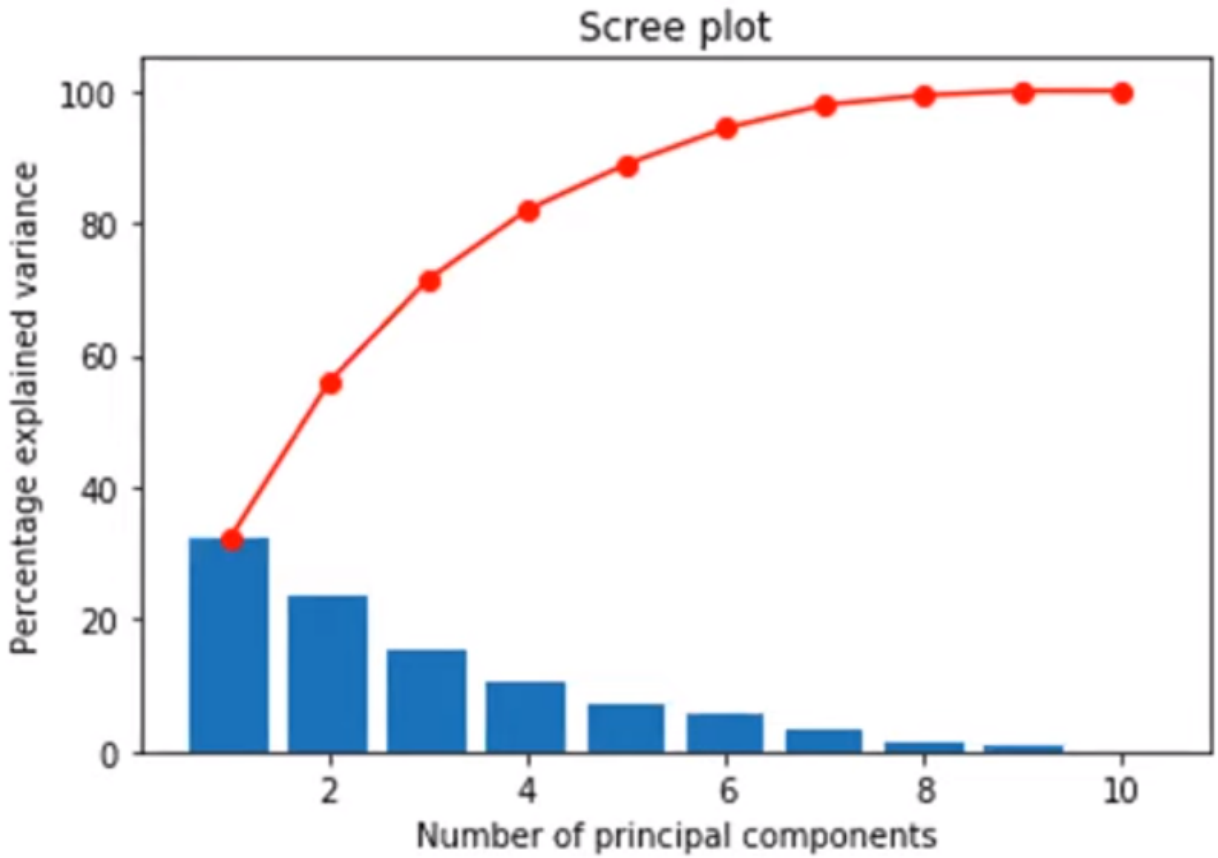
Calculate covariance matrix for *DataAdjust*.

Calculate eigenvalues λ and eigenvectors x for covariance matrix:

- Eigenvalues λ_j are used for calculation of [% of total variance] (V_j) for each component j:

$$V_j = 100 \cdot \frac{\lambda_j}{\sum_{x=1}^{n} \lambda_x} \qquad \sum_{x=1}^{n} \lambda_x = n$$

CHOOSING THE RIGHT NUMBER OF DIMENSSIONS



PCA – STEPS 5 & 6

Choose components – form feature vector

- Eigenvalues λ and eigenvectors **x** are sorted in descending order
- Component with highest λ is principal component
- *Featurevector*=(x_1, … , x_n) where x_i is a column oriented eigenvector. Contains chosen components.

SUMMARY

Principal component analysis (PCA) is a classical statistical method. This linear transform has been widely used in data analysis and data compression.

The principal components (Eigenvectors) for a dataset can directly be extracted from the covariance matrix.

PCA PROSPERITIES

Eigenvectors and eigenvalues can be computed by decomposition module. By sorting the eigenvectors in the order of descending eigenvalues, we can find directions in which the data has the most significant amounts of its variation.

Principal Components Analysis selects a new set of axes for the data.

These axes are selected in decreasing order of variance within the data.

The axes are also perpendicular to each other.

EXAMPLE- IRIS DATA (UCI REPOSITORY)

Attributes: sepal_length, sepal_width, petal_length and petal_width

$$\boldsymbol{Ce} = \boldsymbol{\lambda e}$$

Covariance Matrix

C_{XX}	X_1	X_2	X_3	X_4
X_1	$C_{1,1} = 0.6811$	$C_{1,2} = -0.0390$	$C_{1,3} = 1.2652$	$C_{1,4} = 0.5135$
X_2	$C_{2,1} = -0.0390$	$C_{2,2} = 0.1868$	$C_{2,3} = -0.3196$	$C_{2,4} = -0.1172$
X_3	$C_{3,1} = 1.2652$	$C_{3,2} = -0.3196$	$C_{3,3} = 3.0924$	$C_{3,4} = 1.2877$
X_4	$C_{4,1} = 0.5135$	$C_{4,2} = -0.1172$	$C_{4,3} = 1.2877$	$C_{4,4} = 0.5785$

EXAMPLE--PCA

Eigenvectors

	e_1	e_2	e_3	e_4
sepal_length	0.3616	0.6565	-0.5810	0.3173
sepal_width	-0.0823	0.7297	0.5964	-0.3241
petal_length	0.8566	-0.1758	0.0725	-0.4797
petal_width	0.3588	-0.0747	0.5491	0.7511

Eigenvalues

	Eigenvalue
λ_1	4.22
λ_2	0.24
λ_3	0.08
λ_4	0.02

ADVANTAGES OF PCA

1. **Dimensionality reduction:** By determining the most crucial features or components, PCA reduces the dimensionality of the data, which is one of its primary benefits. This can be helpful when the initial data contains a lot of variables and is therefore challenging to visualize or analyze.

2. **Feature Extraction:** PCA can also be used to derive new features or elements from the original data that might be more insightful or understandable than the original features. This is particularly helpful when the initial features are correlated or noisy.

3. **Data visualization:** By projecting the data onto the first few principal components, PCA can be used to visualize high-dimensional data in two or three dimensions. This can aid in locating data patterns or clusters that may not have been visible in the initial high-dimensional space.

4. **Noise Reduction:** By locating the underlying signal or pattern in the data, PCA can also be used to lessen the impacts of noise or measurement errors in the data.

5. **Multicollinearity:** When two or more variables are strongly correlated, there is multicollinearity in the data, which PCA can handle. PCA can lessen the impacts of multicollinearity on the analysis by identifying the most crucial features or components.

DISADVANTAGES OF PCA

1. **Interpretability**: Although PCA is effective at reducing the dimensionality of data and spotting patterns, the resulting principal components are not always simple to understand or describe in terms of the original features.
2. **Information loss:** PCA involves choosing a subset of the most crucial features or components in order to reduce the dimensionality of the data. While this can be helpful for streamlining the data and lowering noise, if crucial features are not included in the components chosen, information loss may also result.
3. **Outliers:** Because PCA is susceptible to anomalies in the data, the resulting principal components may be significantly impacted. The covariance matrix can be distorted by outliers, which can make it harder to identify the most crucial characteristics.
4. **Scaling:** PCA makes the assumption that the data is scaled and centralized, which can be a drawback in some circumstances.
5. **Computing complexity:** For big datasets, it may be costly to compute the eigenvectors and eigenvalues of the covariance matrix. This may restrict PCA's ability to scale and render it useless for some uses.

QUESTIONS?